PROTOCOL FOR ENCODING MAP ROTATION FOR FORWARD SECRECY

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Title: PROTOCOL FOR ENCODING MAP ROTATION FOR FORWARD SECRECY

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Abstract

A communication protocol that provides forward secrecy at the physical encoding layer by periodically rotating the data encoding map used during a communication session. The method comprises: initiating a session with a first disposable encoding map; monitoring a trigger condition, such as the number of transmitted data frames or elapsed time; upon the trigger condition being met, autonomously generating and switching to a new, second disposable encoding map mid-session; and continuing the communication. Each map is derived from a shared secret, but a compromise of the secret at one point in time does not allow for the decryption of traffic from an earlier epoch that was encoded with a since-discarded map. This prevents retrospective cryptanalysis of captured data streams.

Background of the Invention

Forward secrecy is a critical property of secure communication protocols, ensuring that the compromise of a long-term secret key does not compromise past session keys. This is typically achieved through key exchange protocols like Diffie-Hellman, which generate

ephemeral session keys. In the new paradigm of physical layer security using disposable encoding maps, a similar mechanism is needed. If a single disposable map is used for a very long session, an attacker could potentially capture enough data to perform frequency analysis or other attacks. A protocol is needed to limit the lifetime of any single encoding map to ensure forward secrecy at the encoding layer.

Summary of the Invention

The present invention is a protocol for encoding map rotation that provides forward secrecy for communication systems using disposable encoding maps on high-cardinality channels. The protocol ensures that any single encoding map is used for only a limited period or for a limited amount of data, thwarting cryptanalytic attacks that require large samples of data encoded with the same map.

The method operates as an enhancement to a secure communication session that uses a disposable encoding map derived from a shared secret. The protocol defines a trigger condition for map rotation. This trigger can be a certain number of data frames transmitted, a specific amount of time elapsed, or a combination of both. The communicating endpoints continuously monitor for this trigger.

When the trigger condition is met, both endpoints, which share a secret seed and a deterministic algorithm, can independently and synchronously generate the *next* encoding map in a pre-determined sequence. They then seamlessly switch from the old map to the new map at a designated frame number or time. The old map is then securely discarded. This process repeats throughout the communication session. An attacker who compromises the shared secret at time T would only be able to generate the map for the current epoch (Map N), but would not be able to reconstruct the previous, discarded maps (Map N-1, Map N-2, etc.), and thus cannot decode previously intercepted traffic.

~~Statement of License Grant to the United States Government~~

~~The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental,~~

~~defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.~~

Brief Description of the Drawings

FIG. 1 is a diagram illustrating the process of encoding map rotation during a communication session.

Detailed Description of the Invention

Referring to **FIG. 1**, the protocol is shown in operation over time. A communication session (101) begins between two endpoints using a first disposable map, Map A (102), which is generated from a shared secret. The endpoints transmit a series of data frames (103) encoded with Map A. A trigger condition (104) is defined, for example, a count of 10,000 frames. As the session proceeds, the endpoints count the transmitted frames. When frame 10,000 is reached, the trigger is activated. Both endpoints independently generate the next map in the sequence, Map B (105), and discard Map A. All subsequent data frames (106) are transmitted using Map B, until the next trigger condition is met, at which point they would switch to Map C, and so on. This periodic rotation ensures that any single map is used for only a limited window of communication.

~~Claims~~

~~What is claimed is:~~

1. ~~A method for providing forward secrecy in a communication session between two devices using disposable encoding maps, the method comprising the steps of:~~
    - ~~commencing data transmission for the session using a first data encoding map;~~
    - ~~monitoring for the occurrence of a pre-defined trigger condition;~~
    - ~~upon occurrence of the trigger condition, ceasing to use the first data encoding map and commencing use of a second, different data encoding map for subsequent data transmission in the same session; and~~
    - ~~discarding the first data encoding map.~~
2. ~~The method of claim 1, wherein the trigger condition is the transmission of a predefined number of data frames.~~

3. ~~The method of claim 1, wherein the trigger condition is the expiration of a pre-defined time interval.~~

4. ~~The method of claim 1, wherein the first and second data encoding maps are generated independently by each device based on a shared secret and a deterministic algorithm.~~

5. ~~The method of claim 4, wherein the second map is the next map in a pseudo-randomly generated sequence of maps.~~

6. ~~The method of claim 1, wherein the communication session is conducted over a parallel communication channel with a cardinality greater than 2.~~

7. ~~The method of claim 6, wherein the parallel communication channel is a massively parallel optical interface as described in U.S. Patent Application No. 19/559,705.~~

Drawings



Figure 1: Diagram illustrating the process of encoding map rotation during a communication session.